

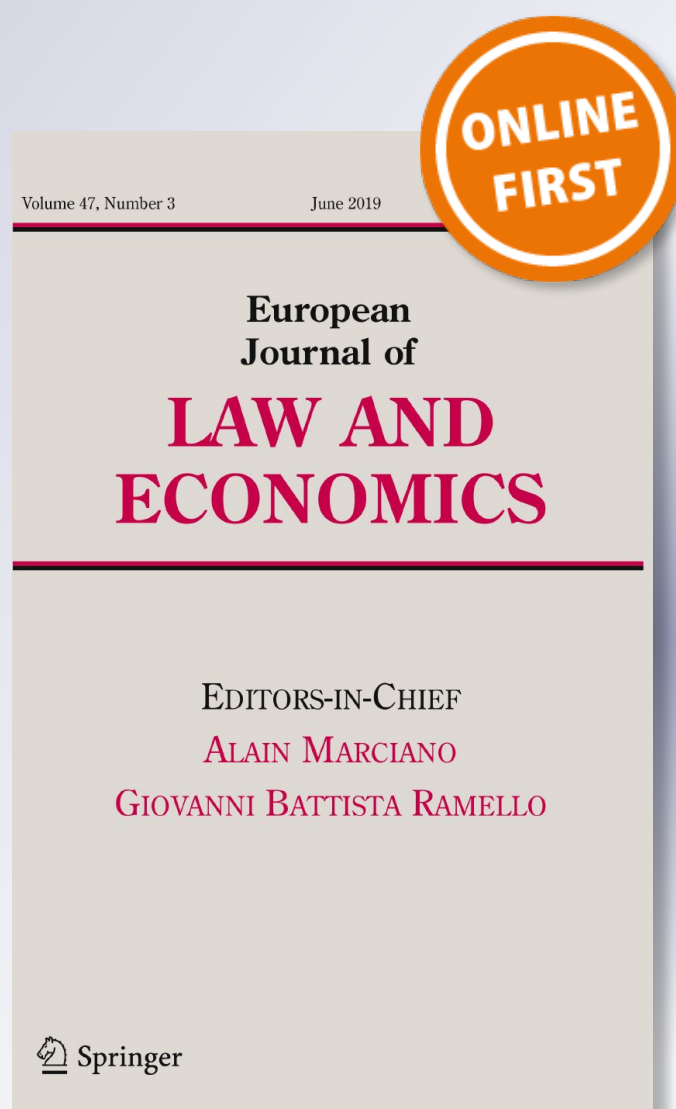
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Indigenous land demarcation conflicts in Brazil: Has the Supreme Court's decision brought (in)stability?

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Abstract

We investigate judiciary decision-making patterns regarding property rights conflicts between native Brazilians and rural farmers in the Midwest region of Brazil. Our main contribution is the use of the qualitative comparative analysis (QCA) method for the evaluation of the issue. We use QCA to examine a unique database composed of cases heard by a regional federal court in Brazil between 1999 and 2013. Our empirical analysis is based on the outcomes of the individual judicial procedures along with the specific laws and jurisprudences evoked by the federal judges in making their decisions. We find evidence that a major case settled by the Brazilian Supreme Court in 2009, celebrated as a landmark in national jurisprudence and expected to bring stability to the conflicts, did not have this effect on lower courts' judgments. In fact, the 2009 decision triggered a proliferation of different interpretations among judges about how to analyze the land conflicts, thus changing the structure of the judges' decisions. Based on these findings, we can shed new light on the dynamics of judicial decision-making regarding property rights conflicts of indigenous lands in Brazil.

Keywords Indigenous · Landowners · Land conflicts · Property rights · Judicial system · QCA

JEL Classification Q15 · J15 · K41 · P48

1 Introduction

Legal conflicts associated with indigenous lands are increasing in frequency in Brazil. The vast majority of the conflicts stem from the demand for the demarcation of indigenous reservations within important productive regions of the country. The

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situation is particularly acute in the Midwest region of Brazil due to a complex legal situation. Based on established legal rules, native Brazilians can claim legal rights to their ancestors' lands. Farmers, on the other hand, have official documents granted by the state stretching back to the first half of the eighteenth century which assure them ownership of their rural properties (Griffiths 2004; Plant and Hvalkof 2001). In response, courts are continuously called upon to settle specific conflicts over property rights.

Within this context, in the early 2000s the Brazilian Federal Supreme Court (STF) heard a major case (the so-called *Raposa Serra do Sol* case) related to the creation of a 1.7-million-hectare indigenous reservation in the northern region of the country. Creating this indigenous reserve involved a long, complex process. The Brazilian National Indian Foundation (FUNAI) proposed the reservation in 1993. After a series of judicial decisions, the Brazilian Federal Supreme Court heard the case. The Court's decision in 2009 was celebrated as a landmark in national jurisprudence and was expected to bring stability to conflicts throughout the country. The underlying reasoning was simple: although Brazil does not have an explicit *stare decisis* principle in adjudication, the Supreme Court Justices expressed the view that the absence of formal binding precedents should not prevent other jurisdictions from following the case law it had established.¹ In the words of the Justices, although the ruling may not have explicit binding effect, "the decision bears the intellectual and persuasive power of the highest court of the country".² Accordingly, the paradigmatic jurisprudence brought by the Brazilian Supreme Court in 2009 was expected to drive lower courts to a more homogeneous understanding of the complex issue of indigenous land demarcation. This reasoning is consistent with the literature on the legal evolution of judicial decisions (e.g., Fon and Parisi 2006).

However, our investigation shows a different phenomenon. We look at a unique database composed of cases of indigenous land conflicts heard by *Tribunal Regional Federal* (Regional Federal Court—2nd instance, appellate level court) between 1999 and 2013. Our empirical analysis is based on the outcomes of individual judicial procedures along with the specific laws and jurisprudences evoked by the federal judges in making their decisions. Our main contribution is the use of qualitative comparative analysis (QCA) (Rihoux and Ragin 2009) for the evaluation of the matter. QCA is different from a traditional econometric approach, as it applies a Boolean algorithm to verify whether one or more combinations of the laws and/or jurisprudences comprise a sufficient or necessary condition to create a specific outcome (i.e. a pro-farmer decision).

We find evidence that the Supreme Court's decision has culminated in an increase in the number of possible configurations that may lead to a particular outcome, meaning that the 2009 decision triggered a proliferation of different understandings amongst the judges on how to analyze the land conflicts. The result is robust

¹ de Oliveira and Garoupa (2012) note that *stare decisis* (as well as *certiorari*) "[responds] to a long debate in the Brazilian legal community on [...] the role of the higher courts in establishing case law" (p. 555).

² See <http://www.stf.jus.br/portal/cms/verNoticiaDetalhe.asp?idConteudo=251738>.

when controlling for the type of plaintiff and when examining the contrasting judicial decision (i.e., pro-Indian decisions). The result is also robust when compared to more ‘traditional’ econometric approaches. We argue that these findings uncover novel evidence about the dynamics of judicial decisions regarding property rights conflicts between native Brazilians and rural farmers.

The rest of our article is structured as follows. First, we present the main aspects of indigenous land demarcation conflicts in Brazil. This is the basic context of our investigation. We then present the data and method used in our analysis and discuss the results of our estimation as well as its robustness. We conclude with a discussion of the main implications of the study, the limitations of our paper and possible avenues of future research.

2 Indigenous land demarcation conflicts in Brazil

According to the UN (2018), the lack of guarantees of land rights is one of the main causes of violations of the rights of indigenous peoples. This issue permeates the history of conquest and colonization of countries or regions, and it would not be different in Brazil. From the efforts of dominance and catechesis of the indigenous population in the sixteenth century to the creation of indigenous reserves in more recent times, conflicts between indigenous and farmers, timber merchants and the infrastructure industry in Brazil remains present. At stake is the dispute over land already approved as indigenous reservations as well as new demands for the demarcation of new reserves based on the ancestral right of the indigenous people for their original territory.

The tangle of laws, decrees, and other legal rules historically created in Brazil shows the fragility of the institutional framework for addressing the subject. Currently, the “Statute of the Indigenous People” (*Estatuto do Índio*; Lei 6.001/1973), the Federal Constitution (1988) and Brazil’s subscription to international treaties as the Universal Declaration of Human Rights, the International Covenant on Civil and Political Rights and the United Nations Declaration on the Rights of Indigenous Peoples are not enough to pacify the issue. More importantly, the effectiveness of laws in Brazil is, at best, partial because of the sometimes-poor implementation of legal regulations. This opens room for continuous advances and setbacks.

2.1 Legal dimension

Although the rights of indigenous people have been officially recognized in Brazil since the first constitutions were enacted in the Republican period (after 1889), only in 1973 a significant body of provisions was consolidated under a common regulation, known as the “Statute of the Indigenous People” (Law 6.001/1973). Specifically, section IX, article 2, states that, “In order to protect indigenous communities and to preserve their rights, [the Statue] will guarantee that they [native Brazilians] have permanent possession of the lands in which they live, granting them exclusive utilization of the natural wealth and all the utilities found in those lands.”

Some years after the enactment of the Statute of the Indigenous People, a new federal constitution was introduced in Brazil. Adopted in 1988, it recognizes “indigenous people and their social organization, their costumes, languages, beliefs and tradition, and also their rights over the land that they traditionally occupy” (Brazilian Federal Constitution, Article 231). As Souza and Barbosa (2011) pose: “[...] indigenous people’s legislation has [...] in each historic period, been impregnated by the models of extermination, [then] of integration, and [finally] of recognition of the indigenous peoples’ rights and expansion of their guarantees”. The Federal Constitution of 1988 represented the last model, revealing an advancement in the legal system that protects indigenous people’s interests and needs.

Specifically, Ferraz Junior (2004) believes the 1988 Constitution reasonably defined what it considers to be land for indigenous demarcation. He cites four defining elements: (a) land on which indigenous people live in permanent manner, (b) that is used for their productive activities, (c) is indispensable for the preservation of the environmental resources needed for their well-being, and (d) is necessary for the physical and cultural reproduction, according to their uses, costumes and traditions. Article 231 of the Constitution is also clear in allocating the duty of indigenous land demarcation to the Union.

Ferraz Junior (2004) further believes that the Federal Constitution of 1988 contains, within itself, the means for solving potential conflicts between private land owners and indigenous people: “Property rights and vested rights—as those of indigenous peoples’ rights on land possession—are *recognized* by the Constitution. It is no use to talk about the prevalence of one over the others. It is discussing the exercise of these rights that potential conflicts have to be examined, in the same manner as one discusses the conflicts that eventually involves fundamental rights (for instances, the conflict between the right to information and the right to privacy)” (p. 698).

Backed on this institutional framework, FUNAI proposed in 1993 the creation of the indigenous reservation of *Raposa Serra do Sol*, located in the state of Roraima in the northern region of Brazil. This 1.7 million hectares indigenous reservation had been inhabited by thousands of Indians of different ethnic groups (Heck et al. 2005). It is also an area of intense economic activities, including livestock and rice production, and has the potential for mineral exploration and tourism development due to its scenic beauty (Lauriola 2003). From 1993 to 2009, a series of judicial decisions and legal challenges made their way through the courts. Eventually, the case achieved enormous visibility, which made the 2009 decision of the Brazilian Supreme Court (STF) a potential landmark decision for similar cases all over the country. The case also illustrates the difficulty of the issue of indigenous land demarcation in Brazil: when it went to trial, it was the lengthiest case ever faced by the STF.

In general terms, the *Raposa Serra do Sol* case involved the discussion of whether the indigenous people’s lands were to be allocated in a continuous manner, or should be scattered throughout the state of Roraima, so to accommodate the existing rice plantations. STF decided that they must remain continuous. However, such a complex case generated many more implications. The STF further ruled about dozens of related issues: the usage of the natural resources found in those lands—either by

indigenous people and by the Federal Government; the role of the Federal Police and the National Army in that region; the role of NGOs in the conservation of resources; the further role of FUNAI in those lands; the rights and duties of researchers and visitors to the *Raposa Serra do Sol* reservation; the imposition of fares by indigenous people to visitors or to passengers; the forbiddance sale and lease of those lands by the indigenous people; the forbiddance of hunting, fishing, harvest, agriculture, and cattle raising; tax related issues, and so on (Camilo 2010).

Because of these features, the case was a landmark by itself (not to mention the wide repercussions in the media at the time). It was expected that the STF decision would, somehow, have impacts on the future directions of the judicial discussions about this theme.

2.2 Conflicts in Brazil's Midwest region

As previously noted, indigenous conflicts in Brazil stem from the demand for the demarcation of indigenous reservations. Generally, there are two opposing groups, with FUNAI, NGOs and public prosecutors on one side, and farmers and rural unions on the other. These conflicts have led not only to legal disputes, but also to the intensification of violence in the countryside. According to the Brazilian Indigenous Missionary Council (CIMI), the state of Mato Grosso do Sul (Midwest region of Brazil) reported 51 homicides of native Brazilians (62.7% of all indigenous homicides in Brazil)³ in 2011. In the previous 8 years, over 250 indigenous people were killed in the state (Rangel 2011).

Despite the severity of the scenario from a social perspective, the economic perspective is also relevant. Ordinances issued by FUNAI in 2008⁴ set in motion the anthropological studies necessary for demarcating indigenous land reservations in the state. These lands cover 26 counties that are also responsible for the largest volume of grain production in the region. According to the Federation of Agriculture and Livestock of Mato Grosso do Sul (Famasul), these counties comprise 21% of the total area of the state, 28% of the population, 25% of local GDP, 37% of exports, 22% of jobs and 30% of the rural production units. The ordinances for anthropological studies in this area fueled a widespread feeling of legal insecurity among farmers. Landowners, even those with legal documents, were subject to evaluation for the purpose of demarcating indigenous lands.

Conflicts in the region have also seriously affected production of sugarcane and soybeans. Decree no. 6961, issued in 2009, approved the *Agro-Ecological Zoning of Sugarcane*, which declares that planting sugarcane in indigenous areas is irregular

³ These data are refuted by the Federation of Agriculture and Livestock of Mato Grosso do Sul (Famasul 2013). According to a survey conducted by the State Secretary of Justice and Public Security, there were 27 homicides against Indians in 2011, and in 18 cases Indians committed the murders. Moreover, in the 7 cases the authors could not identify, the reasons were interpersonal disputes among Indians themselves. In any case, data confirm the degree of violence in the region. Considering the suicide rate of indigenous people in Mato Grosso do Sul—50% of the national number of suicides in 2011 (CIMI 2011)—the severity of the social scenario is obvious.

⁴ Ordinances nos. 788, 789, 790, 791, 792 and 793 from July, 2008.

and disqualifies producers from receiving public funding. As a result, some of the sugar and ethanol mills cut commercial relations with rural producers whose properties are located in areas under the indigenous land demarcation process. Soybean production, a more consolidated culture in the state, occupies a larger extension of production areas, raising issues of water contamination and truck traffic cutting through indigenous lands. Hence, the current legal uncertainty negatively impacts the economy of the state, suppressing the expansion of production and creating uncertainties for future investments.

Within this context, courts are called to settle specific conflicts about property rights. Given its prominence, the 2009 Supreme Court decision was expected to drive lower courts to a more homogeneous understanding of the complex issue of indigenous land demarcation. To examine whether this really was the case, we present our empirical analysis in the following sections.

3 Data and method

We analyze litigation cases heard by the Third Regional Federal Court in Brazil (TRF3), a second-degree, appellate court that handles cases related to land disputes between indigenous people and farmers in the state of Mato Grosso do Sul. Typically, litigations heard by TRF3 are appeals against decisions made in first-degree courts. All cases used in this study are available in the form of digital files on the court's website. By typing the keywords "indigenous people", "land" and "MS" (abbreviation for the state of Mato Grosso do Sul) on the digital database of the court, close to a hundred cases are identified. We selected all cases heard by TRF3 between 1999 and 2013 (60 cases).

To build our database, we conducted an in-depth analysis of each case. We primarily focused on identifying the pattern of the decision, i.e., the law(s) and/or jurisprudence(s) on which the decision was based. There are four main possibilities: (a) the Statute of the Indigenous People; (b) the Federal Constitution, particularly Article 231; (c) Decree N. 1775 of 1996 (*Decreto 1.775/1996*), which establishes the rules under which FUNAI can create study groups and request technical reports for the demarcation of indigenous lands; and (d) the *Raposa Serra do Sol* case, usually referred to as Petition 3388, from the state of Roraima (PET 3388/RR), which has been previously described in details.

Another important dimension of the analysis is the type of litigant, of which there are six: (1) individuals, (2) enterprise or commercial associations, (3) the Federal Union or other public entities, such as FUNAI and the Public Prosecutor's Office (*Ministério Público Federal—MPF*), (4) indigenous people as individuals,⁵ (5) the

⁵ As a usual rule, native Brazilians as individuals are not allowed to be an active party in litigation since the law does not recognize them as legally capable. The Federal Union or FUNAI is their usual representative. Only under very special circumstances are native Brazilians allowed to represent themselves in a legal suit.

state of Matos Grosso do Sul or its counties, and (6) a judicial decision.⁶ We also collected information on the year of court decision (before or after 2009), the party favored by the decision and if the judge's decision was in opposition to the previous decision from the lower court.

In Appendix 1, we provide further details about the methodological steps involved in the selection of the sample analyzed. In Appendix 2, we present a table with the full list of cases used in this paper. All 60 cases are identified by their official ID numbers (as registered by the Federal Court), as well as other complementary information.

3.1 Qualitative comparative analysis

To analyze the pattern of judicial decision-making, we employ qualitative comparative analysis (QCA). QCA is a methodology developed initially in social and political sciences (Rihoux and Ragin 2009). Its objective is to find logical patterns for the presence or absence of each condition included in the model in reaching a specific outcome. The technique is close to a causal model, yet it has an advantage over regular econometric and statistical models by allowing for the examination of combinations that lead to a certain result—i.e., multiple conjunctural causation. This means that the model accounts for the potential presence of several paths to achieve a certain result, i.e., “equifinality” (Campbell et al. 2016). Recent applications of QCA have been performed in different areas ranging from public administration to organization theory and sports management. Andrews et al. (2015), for instance, evidence what are the combinations of attributes in public organizations that lead to the provision of superior public services. Dwivedi et al. (2018) show which firms characteristics jointly lead to women CEO succession. Aversa et al. (2015) assess which aspects within Formula 1 teams lead to successful results in the championship. In our research, we apply QCA to check whether any combination of the laws or jurisprudences giving support to decisions along with the type of the plaintiff/defendant involved in a particular case is a sufficient or necessary condition to create the outcome of interest (i.e., a pro-farmer decision). Again, the idea is to map all possible paths to success or failure and then check which paths achieve the desired outcome (Rihoux and Ragin 2009; Thiem and Dusa 2013).

QCA has diverse forms. The simplest is the crisp set QCA (csQCA), which involves a dichotomous comparison associated, for instance, with the presence or absence of a certain condition (Rihoux and Ragin 2009). However, because csQCA only addresses extremes, we use fuzzy set QCA (fsQCA), which enables us to examine situations where there is variation on the degree of full inclusion or full exclusion of a particular case in a group (Campbell et al. 2016; Fiss 2011; Rihoux and Ragin 2009). Based on our in-depth assessment of the cases, we identified three

⁶ Our sample included six cases in which a judicial decision appeared as the “defendant” party. These were usually situations where plaintiffs appealed against a particular decision, arguing ambiguity or unclearness from the judge. In Brazilian civil procedure, this kind of appeal is called *Embargos de Declaração*.

types of rulings: totally in favor of rural producers, partially in favor of rural producers and totally in favor of native Brazilians. It is important to note that, while QCA addresses mathematical sets theory (Fiss 2007; Ragin 2006), fsQCA uses the notion of a degree by which a particular case is more in or out of a particular set or category (Campbell et al. 2016; Fiss 2011; Ragin 2008).

An additional characteristic of fsQCA is that we can check the influence of distinct types of conditions (and their combinations) on the outcomes (Campbell et al. 2016; Fiss 2011; Judge et al. 2014; Tóth et al. 2015). In other words, we can distinguish between critical conditions—i.e., those we consider the key elements of a given solution—and the conditions that have a marginal influence, despite their contributions to a certain outcome. In the present work, we consider the law or jurisprudence giving support for a specific decision as critical conditions, and the litigant/plaintiff type as marginal condition.

Finally, we have a clear point in time we want to analyze in looking at the patterns of TRF3's decisions: before and after the 2009 decision by the Supreme Court. Therefore, we use a technique to include the temporal dimension in the QCA model (Caren and Panofsky 2005). To do so, we split our sample in two groups. The first group is composed of cases that were decided before the Supreme Court decision (32 cases), and the second group of cases comprises those decided after the Supreme Court decision (28 cases). The basic idea is to compare not only the combination of conditions of the fsQCA models but also consider if the patterns differ longitudinally, i.e., a before and after comparison. The underlying principle is simple: if there is no difference between the patterns leading to pro-farmer decisions before and after the 2009 Supreme Court decision, the decision had no impact on the decision-making process in the lower courts.

3.2 Calibration and measures

Fuzzy set analysis requires an initial calibration of each condition before running the models. Calibration acts as a standard ranking for each condition, taking into account the theoretical framework and the number of occurrences for each condition. The basic idea is to rank all cases according to a fully in—fully out rule (Campbell et al. 2016; Fiss 2011). We have two basic conditions in our QCA model: critical condition (laws and jurisprudence evoked in each decision) and marginal condition (type of plaintiff/defendant).

For the critical condition, we are primarily interested in identifying the presence or absence of specific laws and jurisprudences in a given judicial decision. Accordingly, for each individual decision, we attribute a dichotomous variable indicating the judge's citation of a specific law or jurisprudence in his or her decision. In line with our previous description, there are four dichotomous possibilities (the Indigenous People Statute; the Federal Constitution; Law N. 1776; and PET 44,247), which represent a total of 15 different combinations.

For the marginal condition, plaintiffs and defendants can be classified according to one of the following three 'institutional levels': the micro-level, comprising individual farmers and indigenous people; the meso-level, comprising enterprises or

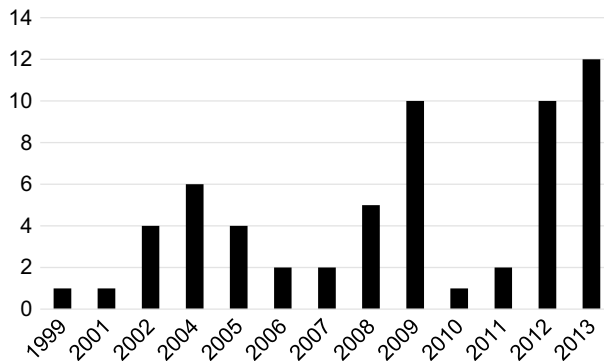


Fig. 1 Cases brought to trial: indigenous land demarcation conflicts, TRF3

commercial associations; and the macro-level, comprising either the federal, state, or municipal government. This categorization seeks to check whether the institutional level of the parties influences the result (Brousseau and Raynaud 2007; North 1990; Williamson 2000).

With this all, it is possible to map all the paths leading to a decision that favors native Brazilians or rural producers.

4 Results

4.1 Descriptive statistics

As indicated in Fig. 1, the number of cases associated with indigenous land demarcation conflicts heard by TRF3 increased between 1999 and 2013. Approximately 67% of the cases were brought to trial from 2008 to 2013, highlighting the rise of the social and economic issues in the state of Mato Grosso do Sul. A significant increase in the number of cases can be noted in 2009 (10 cases), when the Supreme Court issued its decision.

The data also reveals that 46.7% of the first-degree court decisions brought to the second-degree court (TRF3) were overturned. This result is similar to that of an expansive study by Yeung (2019), who measured uncertainty in Brazilian courts by analyzing a sample of 1412 cases related to contractual conflicts over private debts. Their results indicate that 54% of the cases were reformed when going from second-degree to higher courts in Brazil.

Another interesting point is that the state (i.e., federal or local government, municipalities, FUNAI, etc.) became more prevalent as a proponent of legal actions in the period 2008–2013. It seems that institutional agents have become aware of the relevance of land conflict, either in terms of the economic impact on state/municipality tax revenues (caused by the fall in farmers' revenues) or the social impact derived from increasingly violent conflicts between farmers and native Brazilians.

Table 1 Jurisprudence or laws cited in the decisions

	Before 2009 decision (%)	After 2009 decision (%)
Federal constitution	72.0	46.4
Statute of the indigenous people	28.1	39.3
Law N. 1776	40.6	32.1
2009 Supreme Court decision	—	35.1
Other jurisprudence	47.0	57.1

Table 2 Results of cases brought to trial—TRF3

	Before 2009	After 2009	Total
Decisions in favor of farmers	15	14	29
Decisions in favor of native Brazilians	17	14	31
Total	32	28	60

Regarding the legal basis of the decisions, Table 1 describes the frequency of citation of laws and jurisprudences in the decisions issued by TRF3 judges. They cited specific laws, such as the Statute of the Indigenous People, the Federal Constitution or other paradigmatic cases. Interestingly, the Federal Constitution is cited less frequently after the 2009 Supreme Court decision (72% vs. 46.4%). The opposite is true for the Statute of the Indigenous People (28.1% vs. 39.3%) and for the other jurisprudence (47% vs. 57.1%). Most of the time, judges presented more than one legal basis for their decisions (which is why the numbers in each column of the table add up to more than 100%). When analyzing only the cases brought to trial after 2009, the Supreme Court decision is explicitly cited in 35.1% of the cases. However, this number should be read with caution as a citation by the judge does not automatically mean strict adherence or agreement. In fact, it would be hard to believe that a mere reference to the *Raposa Serra do Sol Case* indicates that the TRF3 judge entirely adopted the Supreme Court decision, since cases present their own specificities. Supreme Court jurisprudence is, many times, cited as reference, but not as an indication of alignment by lower court judges.

Of particular interest for our research is the fact that there is no radical change in the trend of court decisions after 2009. As indicated in Table 2, before 2009, 47% of the decisions (15 cases out of a total of 32) were in favor of farmers, compared to 53% of the decisions in favor of indigenous people. After 2009, 50% of the decisions were in favor of farmers.

Moreover, to better visualize the trends in the TRF3 decisions across time comparing before and after the 2009 Supreme Court decision, we non-parametrically estimate the likelihood of favorable decisions for rural producers and native Brazilians having the year 2009 as a discontinuity point. Figure 2 shows a binscatter plot (Stepner 2013) approximation controlled by jurisprudence, plaintiff, and defendant

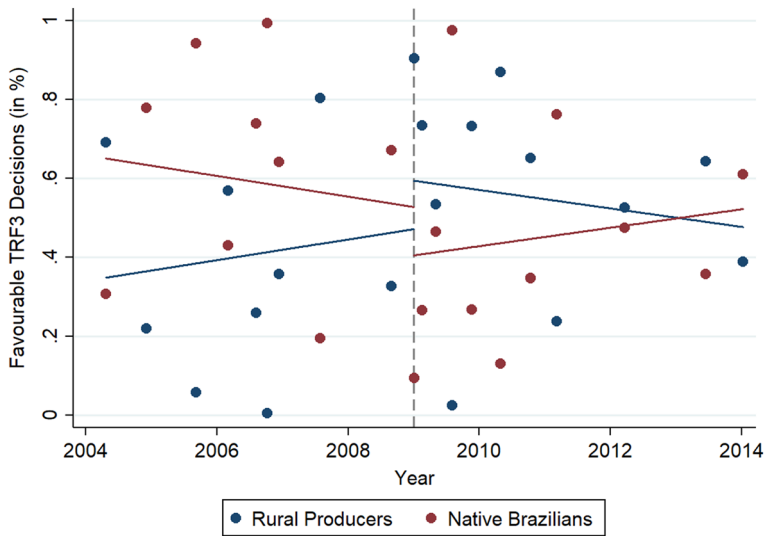


Fig. 2 Trends in TRF3 decisions before and after 2009 STF decision (*Raposa Serra do Sol* case). *Note* Binscatter plot with a non-parametric estimation using 20 bins and discontinuity point on the year 2009 (STF decision over *Raposa Serra do Sol* case). The estimation was performed controlling by jurisprudence, plaintiff and defendant variables

variables. As we can see, despite the slopes of the curves change after 2009, the proportions of cases favorable to either group remained fairly similar throughout time, on average. This may suggest that the 2009 decision of the Supreme Court did not change substantially the resolution of land disputes in favor of either rural producers or indigenous people. However, our question here is of another nature, as we want to understand whether and to what extent the structure of the decisions of TRF3 judges was changed due to the existence of the case law established by the Brazilian Supreme Court in 2009. It is with this intention in mind that we analyze the results of QCA estimates.

4.2 QCA results

We report in Table 3 the QCA results. In this traditional representation, each column in the table represents a possible configuration or ‘solution’. The outcome analyzed here is the judicial decision in favor of rural producers. We examine the structure of the judges’ decisions, taking as a reference the laws and jurisprudences cited in the decision, along with the institutional level of the litigant. Full circles represent laws or jurisprudences that are present and empty circles represent conditions that are absent in a decision in favor of rural producers. Specifically in the case of plaintiffs and defendants, a full circle indicates a macro-level representative (federal government, state government, or municipality), and an empty circle stands for a micro-level representative. It is worth noting that the QCA methodology does not determine a priori the number of possible solutions; this is a result of the estimation

Table 3 TRF3 decisions in favor of rural producers: QCA analysis

Configuration	Before		After					
	2009 decision		2009 decision					
	1A	2A	1B	2B	3B	4B	5B	6B
Federal constitution	○	●	○	●	●	●	○	○
Statute of the indigenous people	○		○	○	●	●	○	●
Law N. 1776	○	●	○	○		○	●	○
2009 Supreme Court decision	○	○	●	○	●	●	○	○
Other jurisprudence	○	●	○	●	●	●	○	●
Plaintiff	●	●	○	●	○		●	
Defendant		○	●	●	●	●		
Consistency ^a	1.000	1.000	1.000	1.000	0.831	1.000	1.000	1.000
Raw coverage ^b	0.267	0.333	0.071	0.143	0.214	0.214	0.143	0.071
Unique coverage	0.267	0.333	0.071	0.143	0.143	0.143	0.143	0.071
Overall solution consistency	1.000		0.948					
Overall solution coverage	0.600		0.786					

○ indicates absence of the law or jurisprudence as well as a micro-level representative; ● indicates presence of the law or jurisprudence as well as a macro-level representative

^aConsistency is the percentage of observations conforming to a specific combination of strategic components with that particular value of the dependent variable or outcome

^bCoverage is the percentage of all observations that exhibit a specific implication or outcome

itself. Thus, the table cannot be read as a common econometric regression (we are not estimating different econometric models in each column). The first column of the table indicates the possible conditions of each configuration, which may or may not be present in a given solution, as indicated by the subsequent columns.

We also report some measures related to estimations. The overall solution coverage refers to the explanatory power of the solutions, i.e., how much of the outcome is covered by all configurations (Rihoux and Ragin 2009). In addition, two measures are available to determine the fit of each configuration (Ragin 2006; Tóth et al. 2015). Consistency measures the extent to which the configuration relates to the outcome (Ragin 2006, 2008). The coverage, on the other hand, assesses the percentage of cases that are associated with a particular configuration, expressing the empirical importance of an identified solution (Fiss 2007). The raw coverage measures the proportion of memberships in the outcome explained by each component of the solution. The unique coverage quantifies the proportion explained only by one solution, excluding memberships that are covered by other solutions (Ragin 2006).

As shown in Table 3, before the 2009 Supreme Court decision, there were two configurations characterizing a judge's decision in favor of rural producers. The first involves a macro-level institution representative as plaintiff, meaning that the judges

decided in favor of the farmer and against the proponent of the case, such as the National Indian Foundation (FUNAI). A second configuration involves the same situation, but the judges explicitly support their decision based on Decree N. 1775, the Federal Constitution, and other minor jurisprudences. These two configurations led to a pro-farmers decision with a consistency of 100% and an overall solution coverage of 60%.

For cases decided after the 2009 Supreme Court decision, the trends are not quite so clear. QCA calculates no less than six different configurations leading to the same outcome (i.e., judicial decision in favor of rural producers). As shown in the combinations of full circles on the right-hand side of Table 3, the conditions are very heterogeneous. Surprisingly, the two 'original' configurations no longer exist after the Supreme Court's decision, with configuration 5B being the only partial exception.

According to solution 1B, one possible configuration involves citing the Supreme Court decision in the absence of any other law or jurisprudence, with the defendant as a macro-level representative. This configuration is the closest way in which a lower court decision in a civil law system can resemble a common law decision. Solutions 2B, 3B and 4B, in contrast, are closer to the civil law tradition, as they involve the federal constitution and the preceding jurisprudence. Solution 2B is based on jurisprudence, with the exception of the 2009 Supreme Court decision, with both litigants as both macro-level representatives. In solutions 3B and 4B, the decisions explicitly cite the 2009 Supreme Court decision and the Statute of the Indigenous People with a macro-level defendant. The difference between them is the absence of Decree N. 1775 in solution 4B. Finally, solution 6B indicates the configuration according to which the decision in favor of rural producers is based on the Statute of Indigenous People and on preceding jurisprudence.

In its totality, these results suggest that after 2009, there is a greater variability in the configurations that rule in favor of farmers. Perhaps more importantly, and in contrast to common belief, the 2009 Supreme Court decision failed to lead to more homogeneity in TRF3 judges' decision-making process.

5 Robustness check

5.1 Cases in favor of indigenous people

To check the robustness of our QCA results, we first use a simple strategy. In line with Bell et al. (2014), we examine the cases in which the judge's decision favors the native people. As QCA is a deterministic, not probabilistic technique, its results do not need to be symmetric (i.e., the paths in favor of farmers might not be the exact opposite of the indigenous people, such as in econometric models). Nevertheless, this test enables us to validate whether the TRF3 has a stable pattern of decision towards one of the parties to the dispute.

Table 4 TRF3 decisions in favor of native Brazilians: QCA analysis

Configuration	Before 2009							After 2009								
	1C	2C	3C	4C	5C	6C	7C	1D	2D	3D	4D	5D	6D	7D	8D	9D
Federal constitution	●	●	●			●	●	○	○	●	●	●	○		●	○
Statute of the indigenous people	○	○	○	○	○	●	●	●	○	○	●	○	○	●	●	○
Law N. 1776	○	○	●	○	●	○		●	○	●	●	○	○		●	○
2009 Supreme Court decision	○	○	○	○	○	○	○	○	○	●	●	●	●	○	○	
Other jurisprudence	○	●	○	●	○	○	○	●	○	○	●	○	●	●	○	●
Plaintiff	●	○	○	○	○	○	○	○	●	●	●			●	●	●
Defendant	○			●	●	●	●	●	○	●	●	○	●	○		●
Consistency ^a	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Raw Coverage ^b	0.059	0.118	0.118	0.118	0.118	0.235	0.118	0.071	0.143	0.071	0.028	0.071	0.071	0.143	0.071	0.099
unique coverage	0.059	0.059	0.059	0.059	0.059	0.176	0.059	0.071	0.143	0.071	0.028	0.071	0.044	0.143	0.071	0.071
Overall solution consistency	1.000							1.000								
Overall solution coverage	0.706							0.742								

○ indicates the absence of the law or jurisprudence as well as a micro-level representative; ● indicates presence of the law or jurisprudence as well as a macro-level representative

^aConsistency is the percentage of observations conforming to a specific combination of strategic components having that particular value of the dependent variable or outcome

^bCoverage is the percentage of all observations that exhibit a specific implication or outcome

Table 4 presents the QCA estimates. Notably, there are many paths that lead to a decision in favor of the indigenous people. Before 2009, there were seven possible configurations that accounted for such decisions. In general, the pattern of the decisions is simple. The Federal Constitution is cited most often, usually in combination with some other law or case law (see solutions 1C, 2C, 3C 6C and 7C). Interestingly, decisions described by configuration 4C come closer to a common law standard in which the judge formats his or her decision based only on case law. After 2009, the number of configurations increases to nine. There are still five configurations that cite the Federal Constitution. Configuration 4D, however, is a more complete pattern in which the judge bases his or her decision not only on the constitution but also on all other dimensions. The pattern of decisions closest to common law also persists. Solution 9D resembles configuration 4C. On the other hand, solution 6D indicates decisions that incorporate the 2009 Supreme Court decision.

The most essential point of this analysis, however, is that the Supreme Court decision did not bring uniformity to decisions of TRF3 judges neither to farmers not to native people. Similar to what happened with decisions favorable to farmers, we find an increase in the number of configurations that led to decisions in favor of native Brazilians after 2009.

5.2 Bootstrap estimations of decisions in favor of rural producers

We also conducted a test to illustrate what would have been our results if we had based our analysis on a more ‘traditional’ probabilistic, econometric approach. As our sample size is fairly small (60 cases) to perform a regular ordinary least squares (OLS) regression, we opted to use a bootstrap technique to evaluate what would have been the result of a regression model if we had a larger sample size.

A basic assumption of the bootstrap method is that the sample examined is representative of the entire population. We argue that this assumption is satisfied in our study since we analyze all cases heard by TRF3 between 1999 and 2013. The bootstrap technique replicates the same sample characteristics R times in order to extrapolate what would be the asymptotic properties of the coefficients for each variable (i.e., condition) of our theoretical model. Bootstrap is adequate for situations where asymptotic inference is difficult to test (Yatchew 2003). We performed a bootstrap with $R = 500$ replications.

The results presented in Table 5 show that no variable/condition is significant in a linear regression model framework when we consider the period after 2009. This result reinforces our main finding from QCA model: the 2009 Supreme Court decision did not generate stability on the following judicial decisions. Also, analogous to our QCA results, the “plaintiff” variable shows a positive significant result ($p < 0.01$) in all models in which it was included before 2009. This result resonates with our QCA model (Table 3) in which “plaintiff” is a necessary condition when we consider the period before 2009. Therefore, the bootstrap estimations help us to assure

Table 5 Bootstrap estimation over TRF3 decisions in favor of rural producers

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Decision for rural producers				Decision for rural producers			
	Before 2009				After 2009			
Plaintiff	0.579** (0.148)		0.759** (0.147)	0.580** (0.195)	−0.168 (0.223)		−0.0303 (0.231)	−0.0325 (0.380)
Defendant		−0.458** (0.156)	0.217 (0.163)	−0.0380 (0.185)		0.356+ (0.190)	0.344 (0.227)	0.335 (0.310)
Federal constitution				−0.608* (0.265)				0.0250 (0.426)
Statute of the indigenous people				0.0924 (0.171)				0.0222 (0.413)
Law N. 1776				0.225 (0.189)				−0.121 (0.326)
2009 Supreme Court decision								0.00393 (0.366)
Other jurisprudence				0.195 (0.191)				0.0875 (0.321)
Constant	0.143 (0.0976)	0.647** (0.114)	−0.0429 (0.136)	0.385 (0.255)	0.611** (0.174)	0.259+ (0.150)	0.287 (0.266)	0.262 (0.425)
Original sample	32	32	32	32	28	28	28	28
Bootstrap sample	500	500	500	500	500	500	500	500
R-squared	0.332	0.194	0.343	0.495	0.021	0.102	0.102	0.125

Robust standard errors in parentheses

** $p < 0.01$; * $p < 0.05$; + $p < 0.1$

the accuracy of the results found previously in the QCA models, although it does not provide us with the configurations associated with a given outcome and it only considers symmetric probabilistic linear relationships.⁷

6 Concluding remarks

In this article, we develop a micro-analytical investigation of judicial decisions concerning indigenous land demarcation conflicts in the Midwest of Brazil. We hypothesize that decisions should have become more homogeneous after 2009,

⁷ Differently from QCA, which is based on deterministic causality, econometric models such as OLS and Bootstrap OLS are based on a probabilistic causality.

with judges evoking similar laws and jurisprudence in their decisions based on the influence exerted by the Supreme Court. Our empirical analysis, however, shows a different phenomenon. The analysis is based on the outcomes of judicial procedures heard by TRF3 between 1999 and 2013, along with the specific laws and jurisprudences evoked by the judges in making their decisions. We use QCA to assess whether one or more combinations of the laws and/or jurisprudences are a sufficient or necessary condition to create a specific outcome (a pro-farmer or pro-indigenous decision).

We find evidence that the Supreme Court's decision has led to an increase in the number of configurations that may lead to a particular outcome. In other words, the 2009 Supreme Court decision did not bring stability to lower courts' judgments (as one might conclude from Table 2); instead, it triggered a proliferation of different interpretations among the judges about how to analyze land conflicts. As a result, it has been more difficult to explain what are exactly the conditions which lead to a pro-farmer or pro-indigenous decision. This tends to raise the level of uncertainty of individuals when they think of going to court. Ultimately, the absence of a clear pattern in judicial decisions could lead to an intensification of disputes in the Brazilian Midwest region, either in the legal arena or in the field (i.e., increasing episodes of violence).

6.1 Limitations

Our paper is not free of limitations. We did not analyze the details of each case that forms our sample. We also did not examine the details of the 2009 Supreme Court decision or the other opinions issued by the Court in subsequent periods. This is a consequence of the method used. The QCA analysis, however, allows us to identify decision patterns, thus serving as a good starting point for further qualitative studies (e.g., content analysis), which could explore the circumstances of each case and the peculiarities of the facts involved. Overall, much remains to be answered relating to this complex matter. By analyzing the temporal distribution of the decisions, one might observe some tendencies, whose causes are yet unknown. Why there has been an increasing number of cases at the TRF3 related to this particular kind of conflicts? Why the divergence in the judicial decisions seems to be increasing? These questions still have to be addressed by future works on this topic.

6.2 Future research

Our conclusion also raises other questions for further research. From a theoretical perspective, our article suggests that there are more subtle aspects that should be examined in the evolutionary dynamics of legal disputes. Our results show that, in the case of indigenous land demarcation conflicts in the Midwest of Brazil, the structure of judicial decisions (i.e., the specific laws and jurisprudences cited in the decision) may follow a more intricate, pluralistic logic. From an empirical perspective, our paper is, to the best of our knowledge, the first to apply QCA methodology to the examination of judicial decisions, and also to the evaluation of the role of the

Supreme Court in the dynamics of lower courts' interpretations. This type of analysis has the potential to shed light on interesting phenomena that can be disregarded by standard econometric approaches. As such, QCA methodology should be used in combination with, and not as an alternative for, more traditional empirical methodologies. This should be viewed as an invitation for further research using QCA and may bring a new impetus for the empirical analysis of law and judicial decisions.

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Appendix 1: Case search methodology

Database and sample

All cases used for the construction of our sample are real litigation cases effectively judged by the Third Regional Federal Court (TRF-3). This is a second-degree, appellate court that deals with the cases in the state of Mato Grosso do Sul. The judicial cases are available in digital file format on the TRF-3 website, on the "TRF-3 Jurisprudence" page (in the original: "Jurisprudência/Consultar Jurisprudência"). For our search, we used the following keywords "indigenous people" (in the original: "indígenas"), "lands" (in the original: "terras") and "MS" (the abbreviation for the state of Mato Grosso do Sul). We further eliminated a few cases that referred to conflicts in the state of São Paulo (the second state covered by TRF-3). We concluded the search for all cases judged until July 20th, 2013. A total of 98 cases fulfilled the above requirements, and from this total, we created a sample of 60 cases for the current study.

Variables

For each case in our sample, we analyzed the following variables:

- *Variable 1:* Was the court's decision in favor of the indigenous people? The results are classified as $x = 1$ (totally favorable); $x = 2$ (partially favorable); $x = 3$ (totally favorable to rural producers and/or totally unfavorable to the indigenous people). In legal terms, $x = 1$ occurs when the decision of an appeal brought by the indigenous people's representatives (FUNAI, Federal Prosecution Service,

the Federal Union, etc.) is “admitted and deferred”, or when one appeal brought to TRF-3 by rural producers or by their representatives is “denied” and/or “not deferred”. Likewise, $x = 3$ occurs when a lawsuit brought by indigenous representatives is not admitted and/or not deferred, or when an claim brought by the farmers or their representatives is deferred. The variable will be classified as $x = 2$ if the decision partially defers the interests of indigenous people, and partly the interests of rural producers.

- *Variable 2*: Did the judge at TRF-3 amended the lower court decision? ($y = 1$ for “yes” and $y = 2$ for “no”).
- *Variable 3*: type of parties involved in the litigation, i.e. who is the plaintiff, and who is the defendant in the case (see description on Sect. 3).

We also incorporate some control variables (not all of them used in the final regression models):

- The Judge-Rapporteur of the case;
- Trial date at the TRF-3;
- The Municipality where the conflict occurred;
- Whether the decision by TRF-3 was unanimous or not;
- Juridical citations by the rapporteurs. As is customary in Brazilian courts, the rapporteur formats his/her decision based on legal norms or previous jurisprudence. For this specific case (indigenous peoples’ land conflicts), citations are mostly related to:
 - The Statute of the Indigenous People;
 - The 1988 Federal Constitution;
 - The Decree N. 1775 of 1996;
 - The *Raposa Serra do Sol* case, ruled by the Federal Supreme Court in 2009.

Appendix 2: Sample analyzed

TRF-3 Case ID	Judge-Rapporteur	Year of TRF-3 decision	Type of plaintiff (principal)	Type of plaintiff (secondary)	Type of defendant (principal)	Type of plaintiff (secondary)	County of origin (of the lawsuit)	Decision	Was the decision unanimous?	Did the decision reform a previous one?
2000.03.00.040816-3	Mauricio Kato	2002	1		3		Campo Grande	1	1	2
2000.60.00.002532-9	Suzana Camargo	2006	3	3	1		Campo Grande	3	1	2
2001.03.99.021064-0	Marcio Mesquita	2013	3		6		Campo Grande	3	1	2
2001.03.99.021064-0	Leonel Ferreira	2012	3	3	1		Campo Grande	3	1	2
2001.60.00.003866-3	Nelton Dos Santos	2012	1		3		N.A.	3	2	1
2001.60.02.001314-3	Marcio Mesquita	2007	3		1		Dourados	3	2	2
2003.03.00.017161-9	Erik Gramstrup	2004	3		1		Campo Grande	1	2	1
2003.03.00.017162-0	Erik Gramstrup	2004	3		1		Campo Grande	1	2	1
2003.03.00.054549-0	Erik Gramstrup	2004	1	3	1		Campo Grande	1	2	1
2003.03.00.054550-7	Erik Gramstrup	2004	1		3		Campo Grande	1	1	1
2003.03.00.057024-7	Erik Gramstrup	2004	1		3		Campo Grande	1	1	2
2003.03.00.057027-7	Erik Gramstrup	2004	1		3		Campo Grande	1	1	2
2003.60.00.005222-0	Nelton Dos Santos	2012	1		3		Aquidauana	3	2	1
2003.60.00.011984-2	Ramza Tartuce	2013	1		3		Aquidauana	3	1	2
2003.60.02.002540-3	Vesna Kolmar	2012	3		1		Dourados	1	1	
2004.03.00.003087-1	Ana Pezarini	2005	3	3	1		Dourados	2	2	1
2004.03.00.003118-8	Antonio Cedinho	2012	3	3	6		Dourados	3	1	2
2004.03.00.003121-8	Antonio Cedinho	2012	3	3	6		Dourados	3	1	2
2004.03.00.066491-4	Ramza Tartuce	2005	3		1		Dourados	1	1	1
2004.60.00.004489-5	Ramza Tartuce	2008	3		1		Campo Grande	3	1	1

TRF-3 Case ID	Judge-Rapporteur	Year of TRF-3 decision	Type of plaintiff (principal)	Type of plaintiff (secondary)	Type of defendant (principal)	Type of plaintiff (secondary)	County of origin (of the lawsuit)	Decision	Was the decision unanimous?	Did the decision reform a previous one?
2004.60.05.001250-6	Valdeci Dos Santos	2009	3	3	1		Navirai	2	1	1
2005.03.00.002956-3	Suzana Camargo	2005	3		1		Campo Grande	1	2	1
2005.03.00.006042-9	Suzana Camargo	2006	4	3	1		Campo Grande	1	2	1
2005.03.00.031889-5	Cecilia Mello	2011	3		6	1	Ponta Pora	3	1	2
2005.03.00.063274-7	Suzana Camargo	2005	3		1		Navirai	1	2	1
2005.03.00.064533-0	Henrique Herkenhoff	2009	3	3	1		Dourados	3	1	2
2005.03.00.075141-4	Ramza Tartuce	2009	3	3	1		Campo Grande	3	1	1
2006.03.00.0003784-9	Antonio Cedenho	2013	1		6	3	Campo Grande	1	1	2
2006.03.00.089251-8	Valdeci Dos Santos	2009	3		1		Ponta Pora	1	1	1
2006.03.00.089688-3	Valdeci Dos Santos	2009	3		1		Ponta Pora	1	1	1
2006.03.00.105849-6	Valdeci Dos Santos	2009	3		3		Ponta Pora	1	1	1
2006.03.00.107230-4	Henrique Herkenhoff	2008	3		1		Campo Grande	3	1	1
2007.03.00.032225-1	Henrique Herkenhoff	2008	3		1		Navirai	3	1	1
2007.03.00.056925-6	Márcio Mesquita	2008	3		1		Navirai	3	1	1
2007.03.00.095897-2	Johonsom Di Salvo	2008	1		3	3	Ponta Pora	1	1	2
2007.03.99.046388-0	Vesna Kolmar	2009	1	5	3		Maracaju	1	1	2
2008.03.00.007690-6	André Nekatschalow	2013	2		3		Navirai	2	1	1
2008.03.00.044546-8	Antonio Cedenho	2013	3	3	1	1	Campo Grande	2	1	1
2008.60.00.007863-1	Henrique Herkenhoff	2010	3	3	5	2	Campo Grande	3	1	2
2008.60.00.012813-0	Ricardo China	2009	2		3		N.A.	2	1	1
2008.60.05.001990-7	José Lunardelli	2013	5		3		Ponta Pora	1	1	2
2008.60.05.001990-7	José Lunardelli	2013	5		3		Ponta Pora	1	1	2
2008.60.06.001056-1	Johonsom Di Salvo	2012	5		3	3	Navirai	1	1	2

TRF-3 Case ID	Judge-Rapporteur	Year of TRF-3 decision	Type of plaintiff (principal)	Type of plaintiff (secondary)	Type of defendant (principal)	Type of plaintiff (secondary)	County of origin (of the lawsuit)	Decision	Was the decision unanimous?	Did the decision reform a previous one?
2009.03.00.003297-0	Baptista Ferreira	2009	1	1	3	3	Campo Grande	3	2	1
2009.03.00.0033617-2	José Lunardelli	2011	1	1	3	3	Campo Grande	3	2	2
2009.60.00.002638-6	Antonio Cedenho	2013	3	3	2		Campo Grande	1	1	1
2010.03.00.035201-1	Desembargador Federal Presidente	2013	3	3	3	1	Ponta Pora	2	2	1
2010.60.05.000156-9	Raquel Perrini	2012	3		5		Ponta Pora	3	1	2
2012.03.00.000072-3	Desembargador Federal Presidente	2013	3		6	1	Navirai	2	2	1
2012.03.00.002512-4	Vesna Kolmar	2013	2		3	3	Dourados	2	1	2
2012.03.00.010076-6	Paulo Domingues	2012	3		1		Ponta Pora	1	1	1
2012.03.00.010911-3	Paulo Domingues	2012	3		1		Ponta Pora	1	1	1
2013.03.00.003543-2	André Nekatschalow	2013	2		3	3	Ponta Pora	1	1	1
90.03.044950-3	Suzana Camargo	2002	1		3		Porto-Murtinho	1	2	2
92.03.038856-7	Andre Nabarrete	2002	1		3		Porto-Murtinho	1	1	2
92.03.039146-0	David Diniz	2001	1		3		Porto-Murtinho	1	1	2
92.03.079229-5	Andre Nabarrete	2002	1		3		Porto-Murtinho	1	1	2
93.03.039001-6	Theotonio Costa	1999	1		3		Porto-Murtinho	1	1	2
93.03.074503-5	Luiz Stefanini	2007	3		1		N.A.	1	1	2
95.03.022717-8	Marcelo Duarte	2009	1		3		Campo Grande	3	1	1
							Campo Grande	2	1	1

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